CECH-TO-DERIVED MAP AND THE HYPERCOVERING SPECTRAL SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1171-1196 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 1198-1264, the complete author proof (65 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_hypercovering.json (pb=2717, stmt 1171-1196, proof 1198-1264).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author omits the functoriality statements in F and K - named bookkeeping; the construction and identifications are in-span.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-cech-spectral-sequence' also exists in stacks-sites-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0133: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1264–1264. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0132: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1264–1264. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0130: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1264–1264. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 012X: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1264–1264. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FNB: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1264–1264. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01GW: sources/stacks-hypercovering/pinned-original-source.tex, source lines 1104–1116. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let X be an object of $\mathcal{C}$. Let K be a hypercovering of X. Let $\mathcal{F}$ be a sheaf of abelian groups on $\mathcal{C}$. There is a map*

$$s(\mathcal{F}(K)) \longrightarrow R\Gamma(X, \mathcal{F})$$

in $D^+(Ab)$ functorial in $\mathcal{F}$, which induces natural transformations

$$\check{H}^i(K, -) \longrightarrow H^i(X, -)$$

as functors $Ab(\mathcal{C}) \to Ab$. Moreover, there is a spectral sequence $(E_r, d_r)_{r \geq 0}$ with

$$E_2^{p,q} = \check{H}^p(K, \underline{H}^q(\mathcal{F}))$$

converging to $H^{p+q}(X, \mathcal{F})$. This spectral sequence is functorial in $\mathcal{F}$ and in the hypercovering K.

Proof. We could prove this by the same method as employed in the corresponding lemma in the chapter on cohomology. Instead let us prove this by a double complex argument.

Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ in the category of abelian sheaves on $\mathcal{C}$. Consider the double complex $A^{\bullet,\bullet}$ with terms

$$A^{p,q} = \mathcal{I}^q(K_p)$$

where the differential $d_1^{p,q} : A^{p,q} \to A^{p+1,q}$ is the one coming from the differential on the complex $s(\mathcal{I}^q(K))$ associated to the cosimplicial abelian group $\mathcal{I}^p(K)$ and the differential $d_2^{p,q} : A^{p,q} \to A^{p,q+1}$ is the one coming from the differential $\mathcal{I}^q \to \mathcal{I}^{q+1}$. Denote $\mathrm{Tot}(A^{\bullet,\bullet})$ the total complex associated to the double complex $A^{\bullet,\bullet}$, see Homology, Section 0FNB. We will use the two spectral sequences $({}'E_r, {}'d_r)$ and $({}''E_r, {}''d_r)$ associated to this double complex, see Homology, Section 012X.

By Lemma 01GW the complexes $s(\mathcal{I}^q(K))$ are acyclic in positive degrees and have H^0 equal to $\mathcal{I}^q(X)$. Hence by Homology, Lemma 0133 the natural map

$$\mathcal{I}^\bullet(X) \longrightarrow \mathrm{Tot}(A^{\bullet,\bullet})$$

is a quasi-isomorphism of complexes of abelian groups. In particular we conclude that $H^n(\mathrm{Tot}(A^{\bullet,\bullet})) = H^n(X, \mathcal{F})$.

The map $s(\mathcal{F}(K)) \longrightarrow R\Gamma(X, \mathcal{F})$ of the lemma is the composition of the map $s(\mathcal{F}(K)) \to \mathrm{Tot}(A^{\bullet,\bullet})$ followed by the inverse of the displayed quasi-isomorphism above. This works because $\mathcal{I}^\bullet(X)$ is a representative of $R\Gamma(X, \mathcal{F})$.

Consider the spectral sequence $({}'E_r, {}'d_r)_{r \geq 0}$. By Homology, Lemma 0130 we see that

$${}'E_2^{p,q} = H_I^p(H_{II}^q(A^{\bullet,\bullet}))$$

In other words, we first take cohomology with respect to d_2 which gives the groups ${}'E_1^{p,q} = \underline{H}^q(\mathcal{F})(K_p)$. Hence it is indeed the case (by the description of the differential ${}'d_1$) that ${}'E_2^{p,q} = \check{H}^p(K, \underline{H}^q(\mathcal{F}))$. By the above and Homology, Lemma 0132 we see that this converges to $H^n(X, \mathcal{F})$ as desired.

We omit the proof of the statements regarding the functoriality of the above constructions in the abelian sheaf $\mathcal{F}$ and the hypercovering K. $\square$